ELLER COLLEGE OF MANAGEMENT
HSLopez School of Business Analytics

ANALYTICS REPORT

TO: BANK OF AMERICA

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SUBJECT: ANALYSIS OF CREDIT CARD DATA

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Introduction

We analyzed a dataset containing over 20,000 Taiwanese credit card customers to better understand the factors that influence both a customer's credit limit and their probability of defaulting. Our goal was to build two regression models: one predicting credit limit using gender, education level, relationship status, and age; and another predicting default probability using average bill amount, average payment amount, age, and gender. Both models used dummy variables to compare customer categories.

The credit limit model showed that females receive higher limits than males, customers with only a high school education receive lower limits than those with a bachelor's degree, and customers with graduate degrees receive higher limits. Single customers also receive significantly lower credit limits than customers in relationships, and older customers tend to receive slightly higher limits. In the default model, higher average bills increased the likelihood of default, while higher average payments reduced it. Older customers had slightly higher default risk, and females had lower default probabilities than males.

Based on these findings, we recommend that the credit card company target older single customers with higher credit lines, as the model suggests their credit limits grow faster with age and may indicate strong repayment behavior. Tailoring credit products to this segment could improve customer profitability.

Data Analysis

In the sections below, we analyze two regression models created for the credit card company: one predicting credit limit and another predicting the probability of default. For each model, we interpret the coefficients, evaluate overall model accuracy using R^2 and standard error, and provide example predictions using the regression equations. These interpretations help identify customer characteristics associated with higher credit limits or higher default risk.

Credit Limit Model

Regression Equation

$\widehat{Credit\ Limit} = 3352.01 + 363.15(Female(d) - 1141.66(High\ School(d)) + 2319.98(Grad\ School(d)) - 2976.21(Single(d)) + 67.00(Single\ (d) * Age) + 51.15(Age)$

R^2 = We are 11.87% of the way toward perfectly predicting credit limit using this model.

Standard Error = Predictions of Credit Limit using this model are off by an average of $4029.25.

Gender = Credit Limit is $363.15 higher for Females than Males, on average all else constant

Highest Level of Education = Credit Limit is $1141.65 lower for customers with high school as their highest level of education than customers with a bachelor's degree, also credit limit is $2319.98 higher for customers with a master's and above compared to customers with a bachelor's degree, on average all else constant.

Relationship Status = Credit limit is $2976.21 lower for single customers than customers in a relationship, on average all else constant.

Relationship Status and Age = For single customers as age increases by 1 year, credit limit increases by $67.00 more than customers in a relationship, on average all else constant.

Age = As age increase by 1 year, credit limit increases by $51.15, on average all else constant

Example Prediction

We were asked to predict the credit limit of a 45-year-old, married, female with a university degree. Here is the equation with numbers plugged in.

$\widehat{Credit\ Limit} = 3352.01 + 363.15(1) - 1141.66(0) + 2319.98(0) - 2976.21(1) + 67.00(45) + 51.15(45) = 6055.7$

This customer would be predicted to be allowed to spend $6055.70 every month.

Chance of Default Model

Regression Equation =

$$p(\widehat{Deafault}) = 0.23 + 6.66E\text{-}06(Avg\ Bill) - 0.0002(Avg\ Payment) + 0.0009(Age) - 0.030(Female(d))$$

R^2 = We are 1.5% of the way toward perfectly predicting the probability of a customer defaulting on their credit card using this model.

Standard Error = Predictions of probability of defaulting using this model are off by an average of 41.2 percentage points.

Avg Bill = As average bill increases by $1, the probability of defaulting increases by 0.0007 percentage points, on average all else constant.

Avg Payment = As average payment increases by $1 the chance of defaulting decreases by 0.017 percentage points, on average all else constant.

Age = As Age increases by 1 year the chance of defaulting increases by 0.09 percentage points, on average all else constant.
Female = The chance of defaulting is 3.0 percentage points lower for females than males, on average all else constant.

Example Prediction =
We were asked to predict the probability of someone who is 35 years old, male, has an average bill amount of $850, and average payments of $760.

$$p(\widehat{Deafault}) = 0.23 + 6.66E\text{-}06(850) - 0.0002(760) + 0.0009(35) - 0.030(0) = 11.5$$

This customer would be predicted to have a 11.5% chance of defaulting.

Recommendations
We recommend that the credit card company target older single customers with higher credit lines. The model shows that as single customers age, their credit limits increase faster than those of customers in relationships. This suggests that older single customers may have stronger repayment behavior and spending patterns. Increasing credit lines or offering tailored credit products to this group could help the company improve profitability.

We recommend taking these interpretations with caution, as both models may be considered inaccurate due to their large standard errors. High standard error values indicate that many of the model's predictions are likely to be far off from the true values, meaning the results should not be heavily relied on for precise decision-making.

Conclusion

Our analysis showed that credit limit is strongly influenced by demographic factors such as gender, education, relationship status, and age, while the probability of default is driven mainly by payment behavior, average bill amounts, and age. The credit limit model had modest predictive power, and the default model had very low accuracy, suggesting that additional variables—such as income, employment type, credit history, and prior delinquencies—may be needed for more reliable predictions. Future steps should include testing additional predictors, exploring nonlinear effects, and potentially building tree-based models to capture complex patterns. Based on the current results, we recommend that the credit card company consider offering higher credit lines to older single customers, as this group appears to have stronger repayment behavior. If you have any questions or would like to discuss these findings further, please feel free to contact me at alappin@arizona.edu.

Technical Appendix

Figure 1 – Credit Limit Regression Output

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.344463625					
R Square	0.118655189					
Adjusted R Square	0.118437412					
Standard Error	4029.247442					
Observations	24289					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	6	53072948613	8845491435	544.8464035	0	
Residual	24282	3.94214E+11	16234834.95			
Total	24288	4.47287E+11				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	3352.013061	186.5760138	17.96593781	1.04976E-71	2986.312565	3717.713557
Female(d)	363.1540987	53.39832193	6.800852265	1.06415E-11	258.4900938	467.8181036
High School (d)	-1141.656309	75.14601525	-15.19250629	6.87015E-52	-1288.947135	-994.365484
Grad School (d)	2319.977708	57.90555691	40.06485443	0	2206.479245	2433.476172
Single (d)	-2976.214637	235.0082018	-12.66430113	1.2178E-36	-3436.84521	-2515.584065
Single (d) * Age	67.00372855	6.398971665	10.47101504	1.33091E-25	54.46134936	79.54610774
AGE	51.15484864	4.442819752	11.51404997	1.34598E-30	42.44664787	59.86304942
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Figure 2 – Default Regression Output

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.123113498					
R Square	0.015156933					
Adjusted R Square	0.014994712					
Standard Error	0.411755032					
Observations	24289					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	4	63.36396778	15.84099194	93.43391413	5.36729E-79	
Residual	24284	4117.162937	0.169542206			
Total	24288	4180.526905				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	0.227000209	0.011481871	19.77031597	2.56307E-86	0.204495034	0.249505383
Avg bill	6.66022E-06	1.34395E-06	4.955718654	7.25481E-07	4.026E-06	9.29443E-06
Avg Payment	-0.000167677	9.2316E-06	-18.16339033	3.07501E-73	-0.000185772	-0.000149583
AGE	0.000883172	0.000290498	3.04020407	0.002366696	0.000313779	0.001452565
Female(d)	-0.029907955	0.005436409	-5.501417162	3.80551E-08	-0.040563653	-0.019252258
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Figure 3 – Original Introduction

We were given data on over 20,000 Taiwanese credit card data. We wanted to be able to predict how much credit limit a customer would have based on Gender, education relationship status and age. We also wanted to know what the probability a customer would default based on average bill, average payment, and age. We ran a regression for both using dummy variables. We found that Credit Limit is $363.15 higher for Females than Males, Credit Limit is 1141.65 lower for customers with high school as their highest level of education than customers with a bachelor's degree, also credit limit is 2319.98 higher for customers with a master's and above, Credit limit is 2976.21 lower for single customers, and As age increase by 1 year, credit limit increases by $51.15. We also found that As average bill increases by $1, the probability of defaulting increases by 0.0007 percentage points As average payment increases by $1 the chance of defaulting decreases by 0.017 percentage points, As Age increases by 1 year the chance of defaulting increases by 0.09 percentage points, and The chance of defaulting is 3.0 percentage points lower for females than males all of that in on average all else constant. We recommend that the credit card company target older single customers with higher credit lines. The model shows that as single customers age, their credit limits increase faster than those of customers in relationships. This suggests that older single customers may have stronger repayment behavior and spending patterns. Increasing credit lines or offering tailored credit products to this group could help the company improve profitability.

Sources Consulted